KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4331 | 4 Credits: 4 | Program Core | Source-grounded

Mechanics of Robotics

Study the official course objectives in the source syllabus.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4331 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4331.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Automation and Robotics
Course code	4331
Course title	Mechanics of Robotics
Semester	4 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 45
Periods per semester	45

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

13 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Study the official course objectives in the source syllabus.

Course outcomes

CO	Official outcome	Study level
CO1	10 Applying machines Discuss the different mechanical drives used for	See official syllabus
CO2	13 Applying power transmission Familiarize with anatomy, specifications and	See official syllabus
CO3	10 Understanding applications of Robots To impart knowledge on the direct and inverse	See official syllabus
CO4	10 Applying kinematics	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2 2
CO3	CO3 2 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Understand the basic knowledge of kinematics of machines	3	As specified
Module 2	Discuss the different mechanical drives used for Power transmission	5	As specified
Module 3	Familiarize with anatomy, specifications and applications of Robots	3	As specified
Module 4	To impart knowledge on the direct and inverse kinematics	2	As specified

MODULE 1

Understand the basic knowledge of kinematics of machines

This module is presented from the official syllabus scope for Mechanics of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand the basic knowledge of kinematics of machines
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

4 Understanding machine elements

4 Understanding machine elements is an official topic in Module 1 of Mechanics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding machine elements using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding machine elements should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding machine elements means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Understanding machine elements □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Understand the concept of cam and follower

Understand the concept of cam and follower is an official topic in Module 1 of Mechanics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand the concept of cam and follower using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand the concept of cam and follower should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand the concept of cam and follower means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Understand the concept of cam and follower [illegible] definition/meaning [illegible]. [illegible], steps, application [illegible]. Technical terms English-[illegible] [illegible].

Discuss the basics of keys, couplings and shafts 3 Applying Basics of mechanisms - Definition and explanation of the terms Kinematic link or element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom- inversion of mechanism – types of constrained motions – classification of Kinematic pairs – types of Kinematic chain – kinematics inversions of 4 bar and slide crank chain – kinematics of motion – velocity and acceleration calculations. (Simple problems) Cams – classifications of cam and followers– motion of followers - cam terminology – displacement diagrams. Keys, coupling, bearings and Shafts – Uses and types – Design for strength of shafts – Simple problems (Elementary idea needed)

Discuss the basics of keys, couplings and shafts 3 Applying Basics of mechanisms - Definition and explanation of the terms Kinematic link or element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom- inversion of mechanism – types of constrained motions – classification of Kinematic pairs – types of Kinematic chain – kinematics inversions of 4 bar and slide crank chain – kinematics of motion – velocity and acceleration calculations. (Simple problems) Cams – classifications of cam and followers– motion of followers - cam terminology – displacement diagrams. Keys, coupling, bearings and Shafts – Uses and types – Design for strength of shafts – Simple problems (Elementary idea needed) is an official topic in Module 1 of Mechanics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss the basics of keys, couplings and shafts 3 Applying Basics of mechanisms - Definition and explanation of the terms Kinematic link or element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom- inversion of mechanism – types of constrained motions – classification of Kinematic pairs – types of Kinematic chain – kinematics inversions of 4 bar and slide crank chain – kinematics of motion – velocity and acceleration calculations. (Simple problems) Cams – classifications of cam and followers– motion of followers - cam terminology – displacement diagrams. Keys, coupling, bearings and Shafts – Uses and types – Design for strength of shafts – Simple problems (Elementary idea needed) using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss the basics of keys, couplings and shafts 3 applying basics of mechanisms - definition and explanation of the terms kinematic link or element, kinematic pair, kinematic chain, mechanism, degrees of freedom- inversion of mechanism – types of constrained motions – classification of kinematic pairs – types of kinematic chain – kinematics inversions of 4 bar and slide crank chain – kinematics of motion – velocity and acceleration calculations. (simple problems) cams – classifications of cam and followers– motion of followers - cam terminology – displacement diagrams. keys, coupling, bearings and shafts – uses and types – design for strength of shafts – simple problems (elementary idea needed) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss the basics of keys, couplings and shafts 3 Applying Basics of mechanisms - Definition and explanation of the terms Kinematic link or element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom- inversion of mechanism – types of constrained motions – classification of Kinematic pairs – types of Kinematic chain – kinematics inversions of 4 bar and slide crank chain – kinematics of motion – velocity and acceleration calculations. (Simple problems) Cams – classifications of cam and followers– motion of followers - cam terminology – displacement diagrams. Keys, coupling, bearings and Shafts – Uses and types – Design for strength of shafts – Simple problems (Elementary idea needed) means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Discuss the basics of keys, couplings and shafts 3 Applying Basics of mechanisms - Definition and explanation of the terms Kinematic link or element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom- inversion of mechanism – types of constrained motions – classification of

Kinematic pairs – types of Kinematic chain – kinematics inversions of 4 bar and slide crank chain – kinematics of motion – velocity and acceleration calculations. (Simple problems) Cams – classifications of cam and followers– motion of followers - cam terminology – displacement diagrams. Keys, coupling, bearings and Shafts – Uses and types – Design for strength of shafts – Simple problems (Elementary idea needed) □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

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Understand the basic knowledge of kinematics of machines

              Understand the basics of mechanisms of
   M1.01                                                        4
Understanding
              machine elements
   M1.02      Understand the concept of cam and follower            3
Understanding
   M1.03      Discuss the basics of keys, couplings and shafts      3
Applying
 Basics of mechanisms - Definition and explanation of the terms Kinematic link or
 element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom- inversion
of
 mechanism – types of constrained motions – classification of Kinematic pairs –
types of
 Kinematic chain – kinematics inversions of 4 bar and slide crank chain –
kinematics of
 motion – velocity and acceleration calculations. (Simple problems)
 Cams – classifications of cam and followers– motion of followers - cam terminology
–
 displacement diagrams.
 Keys, coupling, bearings and Shafts – Uses and types – Design for strength of
shafts –
 Simple problems (Elementary idea needed)
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Discuss the different mechanical drives used for Power transmission

This module is presented from the official syllabus scope for Mechanics of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Discuss the different mechanical drives used for Power transmission
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

3 Understanding transmission Understand the advantages and applications of

3 Understanding transmission Understand the advantages and applications of is an official topic in Module 2 of Mechanics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding transmission Understand the advantages and applications of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding transmission understand the advantages and applications of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding transmission Understand the advantages and applications of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 3 Understanding transmission Understand the advantages and applications of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

2 Understanding different rope and chain drives Impart knowledge about the gears and gear

2 Understanding different rope and chain drives Impart knowledge about the gears and gear is an official topic in Module 2 of Mechanics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding different rope and chain drives Impart knowledge about the gears and gear using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding different rope and chain drives impart knowledge about the gears and gear should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding different rope and chain drives Impart knowledge about the gears and gear means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Understanding different rope and chain drives Impart knowledge about the gears and gear □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

4 Applying trains. Understand the basic concept of gyroscopic

4 Applying trains. Understand the basic concept of gyroscopic is an official topic in Module 2 of Mechanics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying trains. Understand the basic concept of gyroscopic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying trains. understand the basic concept of gyroscopic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying trains. Understand the basic concept of gyroscopic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 4 Applying trains. Understand the basic concept of gyroscopic [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

2 Understanding forces and mechanical vibrations Discuss the concepts of balancing of forces

2 Understanding forces and mechanical vibrations Discuss the concepts of balancing of forces is an official topic in Module 2 of Mechanics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding forces and mechanical vibrations Discuss the concepts of balancing of forces using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding forces and mechanical vibrations discuss the concepts of balancing of forces should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding forces and mechanical vibrations Discuss the concepts of balancing of forces means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Understanding forces and mechanical vibrations Discuss the concepts of balancing of forces □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

2 Applying Mechanical drives: Belts - Types of belt drives – types of belts- Flat belt, circular belt and V-belt – types - velocity ratio – slip – creep – power transmitted by a belt –Problems. Ropes – Types of rope drives – advantages – classification of wire ropes. Chain – Terms used in chain drives -velocity ratio - classification of chains -advantages. Gears and Gear Trains - Functions of gears – advantages and disadvantages of a gear drive – spur gearnomenclature – simple gear drive – velocity ratio –Length of arc of contact- power transmitted – gear trains – simple gear train– compound gear train –reverted gear train – epicyclic gear train -simple problems. Balancing and vibration: Static and dynamic balancing – Rotating masses – Problems- Reciprocating masses (Elementary ideas) – Gyroscopes –Gyroscopic forces and torques – Applications – Mechanical vibrations – classifications – Free and forced vibration- Degrees of freedom – single degree of freedom– Equations of motion – spring mass system -Natural frequency - Simple problems– Types of Damping – Damped vibration– Torsional vibration of shaft – Critical speeds of shafts

2 Applying Mechanical drives: Belts - Types of belt drives – types of belts- Flat belt, circular belt and V-belt – types - velocity ratio – slip – creep – power transmitted by a belt –Problems. Ropes – Types of rope drives – advantages – classification of wire ropes. Chain – Terms used in chain drives -velocity ratio - classification of chains -advantages. Gears and Gear Trains - Functions of gears – advantages and disadvantages of a gear drive – spur gearnomenclature – simple gear drive – velocity ratio –Length of arc of contact- power transmitted – gear trains – simple gear train– compound gear train –reverted gear train – epicyclic gear train -simple problems. Balancing and vibration: Static and dynamic balancing – Rotating masses – Problems- Reciprocating masses (Elementary ideas) – Gyroscopes –Gyroscopic forces and torques – Applications – Mechanical vibrations – classifications – Free and forced vibration- Degrees of freedom – single degree of freedom– Equations of motion – spring mass system -Natural frequency - Simple problems– Types of Damping – Damped vibration– Torsional vibration of shaft – Critical speeds of shafts is an official topic in Module 2 of Mechanics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying Mechanical drives: Belts - Types of belt drives – types of belts- Flat belt, circular belt and V-belt – types - velocity ratio – slip – creep – power transmitted by a belt –Problems. Ropes – Types of rope drives – advantages – classification of wire ropes. Chain – Terms used in chain drives -velocity ratio - classification of chains -advantages. Gears and Gear Trains - Functions of gears – advantages and disadvantages of a gear drive – spur gearnomenclature – simple gear drive – velocity ratio –Length of arc of contact- power transmitted – gear trains – simple gear train– compound gear train –reverted gear train – epicyclic gear train -simple problems. Balancing and vibration: Static and dynamic balancing – Rotating masses – Problems- Reciprocating masses (Elementary ideas) – Gyroscopes –Gyroscopic forces and torques – Applications – Mechanical vibrations – classifications – Free and forced vibration- Degrees of freedom – single degree of freedom– Equations of motion – spring mass system -Natural frequency - Simple problems– Types of Damping – Damped vibration– Torsional vibration of shaft – Critical speeds of shafts using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying mechanical drives: belts - types of belt drives – types of belts- flat belt, circular belt and v-belt – types - velocity ratio – slip – creep – power transmitted by a belt –problems. ropes – types of rope drives – advantages – classification of wire ropes. chain – terms used in chain drives -velocity ratio - classification of chains -advantages. gears and gear trains - functions of gears – advantages and disadvantages of a gear drive – spur gearnomenclature – simple gear drive – velocity ratio –length of arc of contact- power transmitted – gear trains – simple gear train– compound gear train –reverted gear train – epicyclic gear train -simple problems. balancing and vibration: static and dynamic balancing – rotating masses – problems- reciprocating masses (elementary ideas) – gyroscopes –gyroscopic forces and torques – applications – mechanical vibrations – classifications – free and forced vibration- degrees of freedom – single degree of freedom– equations of motion – spring mass system -natural frequency - simple problems– types of damping – damped vibration– torsional vibration of shaft – critical speeds of shafts should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 2 Applying Mechanical drives: Belts - Types of belt drives – types of belts- Flat belt, circular belt and V-belt – types - velocity ratio – slip – creep – power transmitted by a belt –Problems. Ropes – Types of rope drives – advantages – classification of wire ropes. Chain – Terms used in chain drives -velocity ratio -classification of chains -advantages. Gears and Gear Trains - Functions of gears – advantages and disadvantages of a gear drive – spur gearnomenclature – simple gear drive – velocity ratio –Length of arc of contact- power transmitted – gear trains – simple gear train– compound gear train –reverted gear train – epicyclic gear train -simple problems. Balancing and vibration: Static and dynamic balancing – Rotating masses – Problems- Reciprocating masses (Elementary ideas) – Gyroscopes –Gyroscopic forces and torques – Applications – Mechanical vibrations – classifications – Free and forced vibration- Degrees of freedom – single degree of freedom– Equations of motion – spring mass system -Natural frequency - Simple problems– Types of Damping – Damped vibration– Torsional vibration of shaft – Critical speeds of shafts means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Applying Mechanical drives: Belts - Types of belt drives – types of belts- Flat belt, circular belt and V-belt – types - velocity ratio – slip – creep – power transmitted by a belt –Problems. Ropes – Types of rope drives – advantages – classification of wire ropes. Chain – Terms used in chain drives - velocity ratio -classification of chains -advantages. Gears and Gear Trains - Functions of gears – advantages and disadvantages of a gear drive – spur gearnomenclature – simple gear drive – velocity ratio –Length of arc of contact- power transmitted – gear trains – simple gear train– compound gear train –reverted gear train – epicyclic gear train -simple problems. Balancing and vibration: Static and dynamic balancing – Rotating masses – Problems- Reciprocating masses (Elementary ideas) – Gyroscopes –Gyroscopic forces and torques – Applications – Mechanical vibrations – classifications – Free and forced vibration- Degrees of freedom – single degree of freedom– Equations of motion – spring mass system - Natural frequency - Simple problems– Types of Damping – Damped vibration– Torsional vibration of shaft – Critical speeds of shafts □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Discuss the different mechanical drives used for Power transmission
              Discuss different belt drives for power
   M2.01                                                        3
Understanding
              transmission
              Understand the advantages and applications of
   M2.02                                                        2
Understanding
              different rope and chain drives
              Impart knowledge about the gears and gear
   M2.03                                                        4
Applying
              trains.
              Understand the basic concept of gyroscopic
   M2.04                                                        2
Understanding
              forces and mechanical vibrations
            Discuss the concepts of balancing of forces
  M2.05                                                          2
Applying

Mechanical drives: Belts - Types of belt drives – types of belts- Flat belt,
circular belt and
V-belt – types - velocity ratio – slip – creep – power transmitted by a belt –
Problems.
Ropes – Types of rope drives – advantages – classification of wire ropes. Chain –
Terms
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Familiarize with anatomy, specifications and applications of Robots

This module is presented from the official syllabus scope for Mechanics of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Familiarize with anatomy, specifications and applications of Robots
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain different components of robot 4 Understanding Choose appropriate Robotic configuration and

Explain different components of robot 4 Understanding Choose appropriate Robotic configuration and is an official topic in Module 3 of Mechanics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain different components of robot 4 Understanding Choose appropriate Robotic configuration and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain different components of robot 4 understanding choose appropriate robotic configuration and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain different components of robot 4 Understanding Choose appropriate Robotic configuration and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain different components of robot 4 Understanding Choose appropriate Robotic configuration and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Understanding gripper for a particular application Familiar with various design considerations for

4 Understanding gripper for a particular application Familiar with various design considerations for is an official topic in Module 3 of Mechanics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding gripper for a particular application Familiar with various design considerations for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding gripper for a particular application familiar with various design considerations for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding gripper for a particular application Familiar with various design considerations for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding gripper for a particular application Familiar with various design considerations for □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

3 Understanding the selection of a robot or an application Laws of robotics, Progressive advancement in robots, Robot anatomy: links, joint and joint notation scheme, degree of freedom, Robot configurations - PPP, RPP, RRP, RRR. End-effector and Grippers Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Robot considerations for an application- number of axes, work volume, capacity & speed, stroke &reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control.

3 Understanding the selection of a robot or an application Laws of robotics, Progressive advancement in robots, Robot anatomy: links, joint and joint notation scheme, degree of freedom, Robot configurations - PPP, RPP, RRP, RRR. End-effector and Grippers Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Robot considerations for an application- number of axes, work volume, capacity & speed, stroke &reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control. is an official topic in Module 3 of Mechanics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding the selection of a robot or an application Laws of robotics, Progressive advancement in robots, Robot anatomy: links, joint and joint notation scheme, degree of freedom, Robot configurations - PPP, RPP, RRP, RRR. End-effector and Grippers Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Robot considerations for an application- number of axes, work volume, capacity & speed, stroke &reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding the selection of a robot or an application laws of robotics, progressive advancement in robots, robot anatomy: links, joint and joint notation scheme, degree of freedom, robot configurations - ppp, rpp, rrp, rrr. end-effector and grippers classification of end effectors - mechanical grippers, special tools, magnetic grippers, vacuum grippers, adhesive grippers, active and passive grippers, selection and design considerations of grippers in robot. robot considerations for an application- number of axes, work volume, capacity & speed, stroke &reach, repeatability, precision and accuracy, operating environment, point to point control or continuous path control. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding the selection of a robot or an application Laws of robotics, Progressive advancement in robots, Robot anatomy: links, joint and joint notation scheme, degree of freedom, Robot configurations - PPP, RPP, RRP, RRR. End-effector and Grippers Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Robot considerations for an application- number of axes, work volume, capacity & speed, stroke &reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Understanding the selection of a robot or an application Laws of robotics, Progressive advancement in robots, Robot anatomy: links, joint and joint notation scheme, degree of freedom, Robot configurations - PPP, RPP, RRP, RRR. End-effector and Grippers Classification of End effectors -

mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Robot considerations for an application- number of axes, work volume, capacity & speed, stroke &reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Familiarize with anatomy, specifications and applications of Robots

  M3.01      Explain different components of robot                   4
Understanding
             Choose appropriate Robotic configuration and
  M3.02                                                               4
Understanding
             gripper for a particular application
             Familiar with various design considerations for
  M3.03                                                               3
Understanding
             the selection of a robot or an application
Laws of robotics, Progressive advancement in robots, Robot anatomy: links, joint
and
joint notation scheme, degree of freedom, Robot configurations - PPP, RPP, RRP,
RRR.
End-effector and Grippers Classification of End effectors - mechanical grippers,
special
tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive
grippers,
selection and design considerations of grippers in robot. Robot considerations for
an
application- number of axes, work volume, capacity & speed, stroke &reach,
Repeatability, Precision and Accuracy, Operating environment, point to point
control or
continuous path control.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

To impart knowledge on the direct and inverse kinematics

This module is presented from the official syllabus scope for Mechanics of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: To impart knowledge on the direct and inverse kinematics
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

5 inverse kinematics of manipulator. nding Students can able to apply fundamentals of

5 inverse kinematics of manipulator. nding Students can able to apply fundamentals of is an official topic in Module 4 of Mechanics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 inverse kinematics of manipulator. nding Students can able to apply fundamentals of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 inverse kinematics of manipulator. nding students can able to apply fundamentals of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 inverse kinematics of manipulator. nding Students can able to apply fundamentals of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 5 inverse kinematics of manipulator. nding Students can able to apply fundamentals of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

5 Applying mechanism for the design of new mechanisms Introduction to kinematics: Position and orientation of objects, Rotation, Euler angles, Rigid motion representation using Homogenous Transformation matrix. Forward kinematics: Link coordinates, Denavit-Hartenberg Representation, Application of DH convention to different serial kinematic arrangements fitted with spherical wrist. Inverse kinematics – General properties of solutions, Kinematic Decoupling, Inverse kinematic solutions for all basic types of three-link robotic arms fitted with a spherical wrist. Text / Reference: T/R Book Title/Author S.S .Ratan -Theory of Machines - Tata McGraw Hill , New Delhi -2014– 4th T1 edition T2 Sadhu Singh – 'Theory of Machines' – Pearson Education – 2011 – 3rd Edition R1 S S Rao, Mechanical Vibrations, Pearson-2003- 4th edition R2 John. J.Craig, Introduction to Robotics: Mechanics and Control, PHI, 2005. Ashitava Ghoshal, Robotics-Fundamental Concepts and Analysis', Oxford R3 University Press, Sixth impression, 2010.

5 Applying mechanism for the design of new mechanisms Introduction to kinematics: Position and orientation of objects, Rotation, Euler angles, Rigid motion representation using Homogenous Transformation matrix. Forward kinematics: Link coordinates, Denavit-Hartenberg Representation, Application of DH convention to different serial kinematic arrangements fitted with spherical wrist. Inverse kinematics – General properties of solutions, Kinematic Decoupling, Inverse kinematic solutions for all basic types of three-link robotic arms fitted with a spherical wrist. Text / Reference: T/R Book Title/Author S.S .Ratan -Theory of Machines - Tata McGraw Hill , New Delhi -2014– 4th T1 edition T2 Sadhu Singh – 'Theory of Machines' – Pearson Education – 2011 – 3rd Edition R1 S S Rao, Mechanical Vibrations, Pearson-2003- 4th edition R2 John. J.Craig, Introduction to Robotics: Mechanics and Control, PHI, 2005. Ashitava Ghoshal, Robotics-Fundamental Concepts and Analysis', Oxford R3 University Press, Sixth impression, 2010. is an official topic in Module 4 of Mechanics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Applying mechanism for the design of new mechanisms Introduction to kinematics: Position and orientation of objects, Rotation, Euler angles, Rigid motion representation using Homogenous Transformation matrix. Forward kinematics: Link coordinates, Denavit-Hartenberg Representation, Application of DH convention to different serial kinematic arrangements fitted with spherical wrist. Inverse kinematics – General properties of solutions, Kinematic Decoupling, Inverse kinematic solutions for all basic types of three-link robotic arms fitted with a spherical wrist. Text / Reference: T/R Book Title/Author S.S .Ratan -Theory of Machines - Tata McGraw Hill , New Delhi -2014– 4th T1 edition T2 Sadhu Singh – 'Theory of Machines' – Pearson Education – 2011 – 3rd Edition R1 S S Rao, Mechanical Vibrations, Pearson-2003- 4th edition R2 John. J.Craig, Introduction to Robotics: Mechanics and Control, PHI, 2005. Ashitava Ghoshal, Robotics-Fundamental Concepts and Analysis', Oxford R3 University Press, Sixth impression, 2010. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 applying mechanism for the design of new mechanisms introduction to kinematics: position and orientation of objects, rotation, euler angles, rigid motion representation using homogenous transformation matrix. forward kinematics: link coordinates, denavit-hartenberg representation, application of dh convention to different serial kinematic arrangements fitted with spherical wrist. inverse kinematics – general properties of solutions, kinematic decoupling, inverse kinematic solutions for all basic types of three-link robotic arms fitted with a spherical wrist. text / reference: t/r book title/author s.s .ratan -theory of machines - tata mcgraw hill , new delhi -2014– 4th t1 edition t2 sadhu singh – 'theory of machines' – pearson education – 2011 – 3rd edition r1 s s rao, mechanical vibrations, pearson-2003- 4th edition r2 john. j.craig, introduction to robotics: mechanics and control, phi, 2005. ashitava ghoshal, robotics-fundamental concepts and analysis', oxford r3 university press, sixth impression, 2010. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Applying mechanism for the design of new mechanisms Introduction to kinematics: Position and orientation of objects, Rotation, Euler angles, Rigid motion representation using Homogenous Transformation matrix. Forward kinematics: Link coordinates, Denavit-Hartenberg Representation, Application of DH convention to different serial kinematic arrangements fitted with spherical wrist. Inverse kinematics – General properties of solutions, Kinematic Decoupling, Inverse kinematic solutions for all basic types of three-link robotic arms fitted with a spherical

Aspect	What to prepare
	wrist. Text / Reference: T/R Book Title/Author S.S .Ratan -Theory of Machines - Tata McGraw Hill , New Delhi -2014– 4th T1 edition T2 Sadhu Singh – ‘Theory of Machines’ – Pearson Education – 2011 – 3rd Edition R1 S S Rao, Mechanical Vibrations, Pearson-2003- 4th edition R2 John. J.Craig, Introduction to Robotics: Mechanics and Control, PHI, 2005. Ashitava Ghoshal, Robotics-Fundamental Concepts and Analysis’, Oxford R3 University Press, Sixth impression, 2010. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 5 Applying mechanism for the design of new mechanisms Introduction to kinematics: Position and orientation of objects, Rotation, Euler angles, Rigid motion representation using Homogenous Transformation matrix. Forward kinematics: Link coordinates, Denavit-Hartenberg Representation, Application of DH convention to different serial kinematic arrangements fitted with spherical wrist. Inverse kinematics – General properties of solutions, Kinematic Decoupling, Inverse kinematic solutions for all basic types of three-link robotic arms fitted with a spherical wrist. Text / Reference: T/R Book Title/Author S.S .Ratan -Theory of Machines - Tata McGraw Hill , New Delhi -2014– 4th T1 edition T2 Sadhu Singh – ‘Theory of Machines’ – Pearson Education – 2011 – 3rd Edition R1 S S Rao, Mechanical Vibrations, Pearson-2003- 4th edition R2 John. J.Craig, Introduction to Robotics: Mechanics and Control, PHI, 2005. Ashitava Ghoshal, Robotics-Fundamental Concepts and Analysis’, Oxford R3 University Press, Sixth impression, 2010. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
To impart knowledge on the direct and inverse kinematics
            Understand and apply the concept of forward and
Understa
 M4.01                                                                5
            inverse kinematics of manipulator.
nding
            Students can able to apply fundamentals of
 M4.02                                                                5
Applying
            mechanism for the design of new mechanisms

             Series Test – II                                         1
Introduction to kinematics: Position and orientation of objects, Rotation, Euler
angles, Rigid
motion representation using Homogenous Transformation matrix.
Forward kinematics: Link coordinates, Denavit-Hartenberg Representation,
Application of
DH convention to different serial kinematic arrangements fitted with spherical
wrist.
Inverse kinematics – General properties of solutions, Kinematic Decoupling,
Inverse
 kinematic solutions for all basic types of three-link robotic arms fitted with a
spherical
 wrist.

Text / Reference:
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 4 Understanding machine elements. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding machine elements*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Understand the concept of cam and follower. (3 marks)

Answer: Begin with the standard definition or identity of *Understand the concept of cam and follower*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Discuss the basics of keys, couplings and shafts 3 Applying Basics of mechanisms - Definition and explanation of the terms Kinematic link or element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom- inversion of mechanism – types of constrained motions – classification of Kinematic pairs – types of Kinematic chain – kinematics inversions of 4 bar and slide crank chain – kinematics of motion – velocity and acceleration calculations. (Simple problems) Cams – classifications of cam and followers– motion of followers - cam terminology – displacement diagrams. Keys, coupling, bearings and Shafts – Uses and types – Design for strength of shafts – Simple problems (Elementary idea needed). (3 marks)

Answer: Begin with the standard definition or identity of *Discuss the basics of keys, couplings and shafts 3 Applying Basics of mechanisms - Definition and explanation of the terms Kinematic link or element, Kinematic pair, Kinematic chain, mechanism, degrees of freedom- inversion of mechanism – types of constrained motions – classification of Kinematic pairs – types of Kinematic chain – kinematics inversions of 4 bar and slide crank chain – kinematics of motion – velocity and acceleration calculations. (Simple problems) Cams – classifications of cam and followers– motion of followers - cam terminology – displacement diagrams. Keys, coupling, bearings and Shafts – Uses and types – Design for strength of shafts – Simple problems (Elementary idea needed)*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

□□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 2

Define or explain 3 Understanding transmission Understand the advantages and applications of. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding transmission Understand the advantages and applications of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 2 Understanding different rope and chain drives Impart knowledge about the gears and gear. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding different rope and chain drives Impart knowledge about the gears and gear*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Applying trains. Understand the basic concept of gyroscopic. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying trains. Understand the basic concept of gyroscopic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯ definition, ▯▯▯▯▯▯▯ principle/steps, ▯▯▯▯▯▯▯ application ▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯.

Define or explain 2 Understanding forces and mechanical vibrations Discuss the concepts of balancing of forces. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding forces and mechanical vibrations Discuss the concepts of balancing of forces*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯ definition, ▯▯▯▯▯▯▯ principle/steps, ▯▯▯▯▯▯▯ application ▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯.

Module 3

Define or explain Explain different components of robot 4 Understanding Choose appropriate Robotic configuration and. (3 marks)

Answer: Begin with the standard definition or identity of *Explain different components of robot 4 Understanding Choose appropriate Robotic configuration and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯ definition, ▯▯▯▯▯▯▯ principle/steps, ▯▯▯▯▯▯▯ application ▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯.

Define or explain 4 Understanding gripper for a particular application Familiar with various design considerations for. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding gripper for a particular application Familiar with various design considerations for*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding the selection of a robot or an application Laws of robotics, Progressive advancement in robots, Robot anatomy: links, joint and joint notation scheme, degree of freedom, Robot configurations - PPP, RPP, RRP, RRR. End-effector and Grippers Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Robot considerations for an application- number of axes, work volume, capacity & speed, stroke &reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding the selection of a robot or an application Laws of robotics, Progressive advancement in robots, Robot anatomy: links, joint and joint notation scheme, degree of freedom, Robot configurations - PPP, RPP, RRP, RRR. End-effector and Grippers Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Robot considerations for an application- number of axes, work volume, capacity & speed, stroke &reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

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Module 4

Define or explain 5 inverse kinematics of manipulator. nding Students can able to apply fundamentals of. (3 marks)

Answer: Begin with the standard definition or identity of *5 inverse kinematics of manipulator. nding Students can able to apply fundamentals of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Applying mechanism for the design of new mechanisms Introduction to kinematics: Position and orientation of objects, Rotation, Euler angles, Rigid motion representation using Homogenous Transformation matrix. Forward kinematics: Link coordinates, Denavit-Hartenberg Representation, Application of DH convention to different serial kinematic arrangements fitted with spherical wrist. Inverse kinematics – General properties of solutions, Kinematic Decoupling, Inverse kinematic solutions for all basic types of three-link robotic arms fitted with a spherical wrist. Text / Reference: T/R Book Title/Author S.S .Ratan -Theory of Machines - Tata McGraw Hill , New Delhi -2014– 4th T1 edition T2 Sadhu Singh – 'Theory of Machines' – Pearson Education – 2011 – 3rd Edition R1 S S Rao, Mechanical Vibrations, Pearson-2003- 4th edition R2 John. J.Craig, Introduction to Robotics: Mechanics and Control, PHI, 2005. Ashitava Ghoshal, Robotics-Fundamental Concepts and Analysis', Oxford R3 University Press, Sixth impression, 2010.. (3 marks)

Answer: Begin with the standard definition or identity of *5 Applying mechanism for the design of new mechanisms Introduction to kinematics: Position and orientation of objects, Rotation, Euler angles, Rigid motion representation using Homogenous Transformation matrix. Forward kinematics: Link coordinates, Denavit-Hartenberg Representation, Application of DH convention to different serial kinematic arrangements fitted with spherical wrist. Inverse kinematics – General properties of solutions, Kinematic Decoupling, Inverse kinematic solutions for all basic types of three-link robotic arms fitted with a spherical wrist. Text / Reference: T/R Book Title/Author S.S .Ratan -Theory of Machines - Tata McGraw Hill , New Delhi -2014– 4th T1 edition T2 Sadhu Singh – 'Theory of Machines' – Pearson Education – 2011 – 3rd Edition R1 S S Rao, Mechanical Vibrations, Pearson-2003- 4th edition R2 John. J.Craig, Introduction to Robotics: Mechanics and Control, PHI, 2005. Ashitava Ghoshal, Robotics-Fundamental Concepts and Analysis', Oxford R3 University Press, Sixth impression, 2010.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

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PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **4 Understanding machine elements**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **3 Understanding transmission Understand the advantages and applications of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain different components of robot 4 Understanding Choose appropriate Robotic configuration and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **5 inverse kinematics of manipulator. nding Students can able to apply fundamentals of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.

Term	Meaning in this lesson
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

No separate web-resource heading was detected in the extracted text.

IN-PAGE SEARCH

Search this lesson